

WEEK 2: PRODUCT AND CATEGORY MANAGEMENT SYSTEM

Scenario

The e-commerce system maintains information about products and their respective categories. Each product is associated with a category, and the system stores important product details such as product name, price, and stock quantity.

The system also maintains category information including category name and description. This helps organize products into different categories and makes product information easier to manage.

1. Product and Category Tables

Category Table

```
SQL> CREATE TABLE Category(Category_ID VARCHAR2(7),  
 2 Category_Name VARCHAR2(20),  
 3 Description VARCHAR2(50));  
  
Table created.
```

Product Table

```
SQL> CREATE TABLE Product(Product_ID VARCHAR2(7),  
 2 Product_Name VARCHAR2(20),  
 3 Category_ID VARCHAR2(7),  
 4 Price NUMBER(9,2),  
 5 Stock_Quantity Number);  
  
Table created.
```

2. Define Relationships and Constraints

Relationships

The Category and Product tables are related through the **Category_ID** attribute.

One category can contain multiple products. Therefore, the relationship between Category and Product is **One-to-Many (1:M)**.

Primary Key:

In Category Table, Category_ID uniquely identifies each category.

In Product Table, Product_ID uniquely identifies each product.

Foreign Key:

Category_ID acts as Foreign Key in the Product Table.

Other Constraints:

NOT NULL: Product name and category name cannot be empty.

Example: The Product_Name and Category_Name fields should not be left empty because they are required to identify the records.

CHECK: Product price must always be a positive number.

Example: The product price must be greater than zero, so an invalid negative price cannot be entered.

UNIQUE: No two categories can have the same category name.

Example: Every category should have a different Category_Name. Two categories cannot be created with the same name.

Defining relationship and create constraints for product and category:**Primary key:**

```
SQL> ALTER TABLE Category
  2  ADD CONSTRAINT PK_Category PRIMARY KEY (Category_ID);

Table altered.

SQL> ALTER TABLE Product
  2  ADD CONSTRAINT FK_Product_Category
  3  FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID);

Table altered.
```

Creating relationship using Foreign key:

```
SQL> Alter Table Product
  2  Add Constraint FK_Product_Category
  3  FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID);

Table altered.
```

Adding constraints

not null:

```
SQL> Alter Table Category
  2  MODIFY Category_Name VARCHAR(100) NOT NULL;

Table altered.
```

```
SQL> Alter Table Product
  2  MODIFY Product_Name VARCHAR(150) NOT NULL;

Table altered.
```

Check:

```
SQL> ALTER TABLE Product
  2  ADD CONSTRAINT FK_Product_Category
  3  FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID);

Table altered.
```

Unique:

```
SQL> ALTER TABLE Category
  2  ADD CONSTRAINT UQ_Category_Name UNIQUE (Category_Name);

Table altered.
```

3.Inserting values :

Category Table:

```
SQL> INSERT INTO category
  2  VALUES(
  3  '&CATEGORY_ID',
  4  '&CATEGORY_NAME',
  5  '&DESCRIPTION');
Enter value for category_id: C001
old   3: '&CATEGORY_ID',
new   3: 'C001',
Enter value for category_name: ELECTRONICS
old   4: '&CATEGORY_NAME',
new   4: 'ELECTRONICS',
Enter value for description: GADGETS AND DEVICES
old   5: '&DESCRIPTION')
new   5: 'GADGETS AND DEVICES')
```

1 row created.

```
SQL> /
Enter value for category_id: C002
old   3: '&CATEGORY_ID',
new   3: 'C002',
Enter value for category_name: CLOTHING
old   4: '&CATEGORY_NAME',
new   4: 'CLOTHING',
Enter value for description: FASION AND APPAREL
old   5: '&DESCRIPTION')
new   5: 'FASION AND APPAREL')
```

```
SQL> /
Enter value for category_id: C003
old 3: '&CATEGORY_ID',
new 3: 'C003',
Enter value for category_name: BOOKS
old 4: '&CATEGORY_NAME',
new 4: 'BOOKS',
Enter value for description: EDUCATION AND KNOWLEDGE
old 5: '&DESCRIPTION')
new 5: 'EDUCATION AND KNOWLEDGE')
```

1 row created.

```
SQL> /
Enter value for category_id: C004
old 3: '&CATEGORY_ID',
new 3: 'C004',
Enter value for category_name: SPORTS
old 4: '&CATEGORY_NAME',
new 4: 'SPORTS',
Enter value for description: SPORTS EQUIPMENT AND ACCESSORIES
old 5: '&DESCRIPTION')
new 5: 'SPORTS EQUIPMENT AND ACCESSORIES')
```

```

SQL> /
Enter value for category_id: C005
old 3: '&CATEGORY_ID',
new 3: 'C005',
Enter value for category_name: BEAUTY
old 4: '&CATEGORY_NAME',
new 4: 'BEAUTY',
Enter value for description: BEAUTY AND PERSONAL CARE PRODUCTS
old 5: '&DESCRIPTION')
new 5: 'BEAUTY AND PERSONAL CARE PRODUCTS')

1 row created.

SQL> /
Enter value for category_id: C006
old 3: '&CATEGORY_ID',
new 3: 'C006',
Enter value for category_name: GROCERIES
old 4: '&CATEGORY_NAME',
new 4: 'GROCERIES',
Enter value for description: FOOD AND DAILY ESSENTIALS
old 5: '&DESCRIPTION')
new 5: 'FOOD AND DAILY ESSENTIALS')

1 row created.

SQL> /
Enter value for category_id: C007
old 3: '&CATEGORY_ID',
new 3: 'C007',
Enter value for category_name: STATIONARY
old 4: '&CATEGORY_NAME',
new 4: 'STATIONARY',
Enter value for description: SCHOOL AND OFFICE SUPPLIES
old 5: '&DESCRIPTION')
new 5: 'SCHOOL AND OFFICE SUPPLIES')

1 row created.

```

Product table:

```
SQL> INSERT INTO Product
2 VALUES(
3 '&ProductID',
4 '&Product_Name',
5 '&CategoryID',
6 &Price,
7 &Stock_Quantity);
Enter value for productid: P001
old 3: '&ProductID',
new 3: 'P001',
Enter value for product_name: KEYBOARD
old 4: '&Product_Name',
new 4: 'KEYBOARD',
Enter value for categoryid: C001
old 5: '&CategoryID',
new 5: 'C001',
Enter value for price: 2500
old 6: &Price,
new 6: 2500,
Enter value for stock_quantity: 7
old 7: &Stock_Quantity)
new 7: 7)

1 row created.
```

```
SQL> /
Enter value for productid: P004
old 3: '&ProductID',
new 3: 'P004',
Enter value for product_name: HEADPHONES
old 4: '&Product_Name',
new 4: 'HEADPHONES',
Enter value for categoryid: C004
old 5: '&CategoryID',
new 5: 'C004',
Enter value for price: 2000
old 6: &Price,
new 6: 2000,
Enter value for stock_quantity: 9
old 7: &Stock_Quantity)
new 7: 9)

1 row created.
```

```
SQL>
SQL> /
Enter value for productid: P005
old 3: '&ProductID',
new 3: 'P005',
Enter value for product_name: SNEAKERS
old 4: '&Product_Name',
new 4: 'SNEAKERS',
Enter value for categoryid: C006
old 5: '&CategoryID',
new 5: 'C006',
Enter value for price: 2350
old 6: &Price,
new 6: 2350,
Enter value for stock_quantity: 10
old 7: &Stock_Quantity)
new 7: 10)

1 row created.
```

```
SQL> /
Enter value for productid: P004
old 3: '&ProductID',
new 3: 'P004',
Enter value for product_name: HEADPHONES
old 4: '&Product_Name',
new 4: 'HEADPHONES',
Enter value for categoryid: C004
old 5: '&CategoryID',
new 5: 'C004',
Enter value for price: 2000
old 6: &Price,
new 6: 2000,
Enter value for stock_quantity: 9
old 7: &Stock_Quantity)
new 7: 9)

1 row created.
```

```
SQL>
SQL> /
Enter value for productid: P005
old 3: '&ProductID',
new 3: 'P005',
Enter value for product_name: SNEAKERS
old 4: '&Product_Name',
new 4: 'SNEAKERS',
Enter value for categoryid: C006
old 5: '&CategoryID',
new 5: 'C006',
Enter value for price: 2350
old 6: &Price,
new 6: 2350,
Enter value for stock_quantity: 10
old 7: &Stock_Quantity)
new 7: 10)

1 row created.
```

```
SQL> /
Enter value for productid: P006
old 3: '&ProductID',
new 3: 'P006',
Enter value for product_name: BLENDER
old 4: '&Product_Name',
new 4: 'BLENDER',
Enter value for categoryid: C007
old 5: '&CategoryID',
new 5: 'C007',
Enter value for price: 3200
old 6: &Price,
new 6: 3200,
Enter value for stock_quantity: 18
old 7: &Stock_Quantity)
new 7: 18)
```

1 row created.

```
SQL> /
Enter value for productid: P007
old 3: '&ProductID',
new 3: 'P007',
Enter value for product_name: OFFICE CHAIR
old 4: '&Product_Name',
new 4: 'OFFICE CHAIR',
Enter value for categoryid: C007
old 5: '&CategoryID',
new 5: 'C007',
Enter value for price: 5500
old 6: &Price,
new 6: 5500,
Enter value for stock_quantity: 30
old 7: &Stock_Quantity)
new 7: 30)
```

1 row created.

After insertion category table:

```
SQL> SELECT * FROM category;
```

```
CATEGOR
```

```
-----
```

```
CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
C001
```

```
ELECTRONICS
```

```
GADGETS AND DEVICES
```

```
C002
```

```
CLOTHING
```

```
FASHION AND APPAREL
```

```
CATEGOR
```

```
-----
```

```
CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
C003
```

```
BOOKS
```

```
EDUCATION AND KNOWLEDGE
```

```
C004
```

```
SPORTS
```

```
CATEGOR
```

```
-----
```

```
CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
SPORTS EQUIPMENT AND ACCESSORIES
```

```
C005
```

```
BEAUTY
```

```
BEAUTY AND PERSONAL CARE PRODUCTS
```

```
C006
```

```
CATEGOR
```

```
-----
```

```
CATEGORY_NAME
```

```
-----
```

```
DESCRIPTION
```

```
-----
```

```
GROCERIES
```

```
FOOD AND DAILY ESSENTIALS
```

```
C007
```

```
STATIONARY
```

```
SCHOOL AND OFFICE SUPPLIES
```

```
7 rows selected.
```

Aftet insertion product table:

```
SQL> SELECT * FROM product;
```

```
PRODUCT
```

```
-----  
PRODUCT_NAME
```

```
-----  
CATEGOR      PRICE STOCK_QUANTITY
```

```
-----  
P001
```

```
KEYBOARD
```

```
C001
```

```
2500
```

```
7
```

```
P002
```

```
MOUSE
```

```
C002
```

```
800
```

```
15
```

```
PRODUCT
```

```
-----  
PRODUCT_NAME
```

```
-----  
CATEGOR      PRICE STOCK_QUANTITY
```

```
-----  
P003
```

```
PEN SET
```

```
C003
```

```
200
```

```
7
```

```
P004
```

```
HEADPHONES
```

```
PRODUCT
```

```
-----  
PRODUCT_NAME
```

```
-----  
CATEGOR      PRICE STOCK_QUANTITY
```

```
-----  
C004
```

```
2000
```

```
9
```

```
P005
```

```
SNEAKERS
```

```
C006
```

```
2350
```

```
10
```

```
P006
```

```
PRODUCT
```

```
-----  
PRODUCT_NAME
```

```
-----  
CATEGOR      PRICE STOCK_QUANTITY
```

```
-----  
BLENDER
```

```
C007
```

```
3200
```

```
18
```

```
P007
```

```
OFFICE CHAIR
```

```
C007
```

```
5500
```

```
30
```

```
7 rows selected.
```

Update price:

```
SQL> UPDATE Product SET Price = 76000 WHERE Product_Name = 'Tiles';  
1 row updated.
```

Delete a product:

```
SQL> DELETE FROM Product WHERE Product_Name = 'Tiles';  
1 row deleted.
```

4. Category-Wise Product Reports:

SQL query:

```
SELECT c.CATEGORY_ID,  
       c.CATEGORY_NAME,  
       p.PRODUCT_ID,  
       p.PRODUCT_NAME,  
       p.PRICE,  
       p.STOCK_QUANTITY  
FROM product p  
JOIN Category c  
ON p.CATEGORY_ID = c.CATEGORY_ID;
```

```
SQL> SELECT c.Category_Name,p.Product_Name,p.Price,p.Stock_Quantity
2 FROM Product p JOIN Category c ON p.Category_ID = c.Category_ID;
```

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

Electronics

Laptops

67500 2

Clothing

Croptops and shirts

45000 456

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

Stationary

Fish tank

3400 4

Home Appliances

Television

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

75000 5

Mobile Parts

Unruled notes

7500 765

Clothing

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

Tablet

10000 10

6 rows selected.

Counting Number of products in each Category:

```
SQL> SELECT c.Category_Name, COUNT(p.Product_ID) AS Product_Count
  2 FROM Product p JOIN Category c ON p.Category_ID = c.Category_ID
  3 GROUP BY c.Category_Name;
```

CATEGORY_NAME

PRODUCT_COUNT

Clothing 2

Stationary 1

Home Appliances 1

CATEGORY_NAME

PRODUCT_COUNT

Mobile Parts 1

Electronics 1

Finding highest priced product in each category:

CATEGORY_NAME

```
SQL:
  2 PRODUCT_NAME
  3
  4 HIGHEST_PRICE
  5
  6 75000
  7
```

Mobile Parts

CATI Unruled notes
----- 7500

PRODUCT_NAME

HIGHEST_PRICE

Electronics

Laptops 67500

Clothing

Croptops and shirts
45000

CATEGORY_NAME

PRODUCT_NAME

HIGHEST_PRICE

Stationary

Fish tank 3400

Home Appliances

Television

CATEGORY_NAME

Display categories having more than 5 products

```
SQL> SELECT c.Category_Name
  2  FROM Product p JOIN Category c ON p.Category_ID = c.Category_ID
  3  GROUP BY c.Category_Name HAVING COUNT(p.Product_ID) > 5;

no rows selected
```

No rows selected because all categories having less than 5 products.

Find average product price category-wise

```
SQL> SELECT c.Category_Name, AVG(p.Price) AS Avg_Price
  2  FROM Product p JOIN Category c ON p.Category_ID = c.Category_ID
  3  GROUP BY c.Category_Name;
```

CATEGORY_NAME

AVG_PRICE

Clothing
27500

Stationary
3400

Home Appliances
75000

CATEGORY_NAME

AVG_PRICE

Mobile Parts
7500

Electronics
67500